for the economic and political system it was worth it. Mathematical skills are more important to the economy than smelling flowers or dreaming about fairies. For similar reasons, it is likely that future upgrades to the human mind will reflect political needs and market forces.

For example, the US army's 'attention helmet' is meant to help people focus on well-defined tasks and speed up their decision-making process. It may, however, reduce their ability to show empathy and tolerate doubts and inner conflicts. Humanist psychologists have pointed out that people in distress often don't want a quick fix – they want somebody to listen to them and sympathise with their fears and misgivings. Suppose you are having an ongoing crisis in your workplace, because your new boss doesn't appreciate your views, and insists on doing everything her way. After one particularly unhappy day, you pick up the phone and call a friend. But the friend has little time and energy for you, so he cuts you short, and tries to solve your problem: 'Okay. I get it. Well, you really have just two options here: either quit the job, or stay and do what the boss wants. And if I were you, I would quit.' That would hardly help. A really good friend will have patience, and will not be quick to find a solution. He will listen to your distress, and will provide time and space for all your contradictory emotions and gnawing anxieties to surface.

The attention helmet works a bit like the impatient friend. Of course sometimes – on the battlefield, for instance – people need to take firm decisions quickly. But there is more to life than that. If we start using the attention helmet in more and more situations, we may end up losing our ability to tolerate confusion, doubts and contradictions, just as we have lost our ability to smell, dream and pay attention. The system may push us in that direction, because it usually rewards us for the decisions we make rather than for our doubts. Yet a life of resolute decisions and quick fixes may be poorer and shallower than one of doubts and contradictions.

When you mix a practical ability to engineer minds with our ignorance of the mental spectrum and with the narrow interests of governments, armies and corporations, you get a recipe for trouble. We may successfully upgrade our bodies and our brains, while losing our minds in the process. Indeed, techno-humanism